

1 **Assessing the potential of soil cyanobacteria for simultaneous wastewater**
2 **treatment and carbohydrate-enriched biomass production**

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Abstract

In this study, a mixed nitrogen-fixing soil cyanobacterial culture was tested to treat municipal wastewater and produce total carbohydrates in a one-stage operation. Four photobioreactors were operated in semi-continuous mode for 30 days, evaluating the effect of different hydraulic retention times and nutrients and carbon loads on nutrients and organic matter removal, biomass composition, and carbohydrates production. One photobioreactor operated at a hydraulic retention time of 10 d with diluted 2:1 wastewater with distilled water, and other photobioreactors worked at 10 d, 8 d, and 6 d of hydraulic retention times with undiluted sewage. The results evidenced that high hydraulic retention time and low nutrients load achieved the highest removal efficiencies in total nitrogen >95%, total phosphorus 35-78%, total organic carbon >93%, and total inorganic carbon >82%. These high removals led to nitrogen limitation that stimulated a continuous carbohydrates accumulation of up to 48%. Also, a biomass production of 0.05-0.07 mg L⁻¹ d⁻¹ dominated by easy-settling flocs of filamentous nitrogen-fixing cyanobacteria was achieved. Otherwise, lower hydraulic retention time and thus high nutrients load promoted carbon depletion, which led to lower nitrogen and phosphorus removals, low biomass production and contamination of green algae species. The results of this study highlight the nutritional and operational mechanisms of soil microorganisms and strategies to simultaneously clean waste streams and produce valuable by-products.

Keywords: biofuels; green algae; soil crusts; microbial dynamics; closed-photobioreactors.

1. Introduction

Cyanobacteria are a diverse group of prokaryotic microorganisms with significant ecological and physiological variables, not only exhibiting morphological diversity but a

complex metabolism [1]. As the most cyanobacteria species live in the aquatic environment, many others can live in soil and other terrestrial habitats, where they are essential in the functional processes of ecosystems and the cycles of nutrient elements [2,3]. An important contribution of them is in remediation processes preventing the expansion of desertification in soils [4], in this case, they have improved nutrient mineralization, changing the soil surface properties by trapping and keeping water in sandy soils, reducing water infiltration, protecting the soil from erosion, providing fertility, and soil carbon (C) and nitrogen (N) cycling [5–7]. The capacity of cyanobacteria to grow in several environments is attributed to their low nutritional requirements and their high adaptability to changes in the environment. For example, they survive to large variations in light, nutrients starvation, temperature, moisture, and humidity changes [8,9]. In such cases, carbohydrates accumulation is essential to this adaptability, allowing cyanobacteria to create microenvironments within the soil where they can survive [1]. The potential carbohydrates storage in soil cyanobacteria has been exploited to facilitate soil aggregation and stability and improve soil water infiltration. However, the great potential as a source of fine chemicals, biological fertilizers, fuels, and renewable products has been barely explored [10,11].

Cyanobacterial carbohydrates production has become an attractive research area in the last decade because of its potential application as a biofuel substrate [12–14]. These microorganisms have many advantages over higher plants because of producing first and second-generation bioethanol. For instance, they present a fast growth, and the capability of growing under hindering conditions, including in wastewater and short harvesting cycles (~1–10 days) [15]. Although this alternative offers several advantages for biofuels

production, their potential is still limited by several factors, including cost-effective biomass cultivation and to ensure a high amount of carbohydrates [16].

An alternative approach to avoid the high production costs of cyanobacteria that impact biofuels' economic viability is the cultivation of mixed cultures of cyanobacteria in wastewater [17]. With this approach, the production costs associated with pure culture control, nutrients, and water supply would be reduced. However, cyanobacteria cultivation in such a variable media implies certain disadvantages related to the competition with other microorganisms, especially with green microalgae. This may imply that the yields of intracellular compounds can often be reduced with the incidence of other microorganisms [18]. Wastewater treatment has been tested with many cyanobacteria species such as *Arthorspira* sp. [19–21], *Oscillatoria* sp. [22,23], *Phormidium* sp. [24–26] and *Nostoc* sp. [27]. These studies revealed that cyanobacteria possess high wastewater treatment efficiency because of their photoautotrophic nature and their ability to use nitrogen and phosphorus from wastewater [10]. However, most of the studies did not pay attention to factors that may affect polymers production. Only few works focus on the analysis of polymers during this process [28,29]. Thus, aspects concerning operational conditions and nutrients dynamics to maximize polymers production in these systems still need to be addressed.

In this study, we proposed the cultivation of mixed soil crusts dominated by cyanobacteria in a one-stage operation to produce carbohydrates while treating municipal wastewater. To the authors' knowledge, this is the first time that a mixed soil culture is employed to produce valuable polymers within a wastewater treatment process. In the present study, a soil cyanobacterial consortium was cultivated in four semi-continuous closed

photoreactors testing the effect of different hydraulic retention times on nutrients and organic matter removal, biomass composition, and carbohydrates production.

2. Materials and methods

2.1 Inoculum

Cyanobacteria obtained from dry soil crusts were used as inoculum. Dry soil crusts (Appendix A1) were taken from a wastewater treatment plant in Saint-Étienne-de-Tulmont, France. The dry biomass was first hydrated and agitated in an Erlenmeyer with BG-11 growth medium for two days until the cells were suspended. The sample consisted of a mixed culture of nitrogen-fixing cyanobacteria, mostly dominated by two strains resembling species of *Nostoc* sp. and *Tolypothrix* sp., with some cells of heteropolar-shaped *Calothrix* sp. (Fig. 1), identified according to identifications keys of [30,31]. This two species are among the species mostly found in terrestrial environments [2,5,30,32,33]. Suspended biomass was subsequently introduced in a 30 L agitated closed photobioreactor that worked as a stock photobioreactor. The photobioreactor was maintained with nitrogen-free BG-11 growth medium and air-sparing, to maintain a nitrogen-fixing dominated culture during the experiment. The growth medium was changed twice a week by settling the biomass for 30 minutes and manually retiring 28 L of supernatant. The culture in the photobioreactors was maintained in alternate light: dark periods of 12 h. Illumination during the light phase was supplied by a 600 W external metal halide lamp equipped with a digital ballast (model 5500k, Sunmaster, USA) placed at a 70 cm distance from the photobioreactor. This lamp provided approximately 14,500 lx ($204 \mu\text{mol m}^{-2} \text{s}^{-1}$) [34], which corresponds to the irradiance recommended to enhance algal activity (200–400 $\mu\text{mol m}^{-2} \text{s}^{-1}$) [35,36]. Due to the lamp type (cold blue light), culture temperature was minimally influenced during the experimental

time, ranging from 16 to 20 °C throughout the study, and varying around 1-2 °C between light: dark cycles. Agitation consisted of a rotary paddle placed in the middle of the cylinder, rotating at 40 rpm. The pH of the culture was controlled continuously through CO₂ (100% v) (Carbueros Metálicos, Spain) automatic injection at a flow rate of 0.3 L min⁻¹ and a pressure of 0.3–0.5 MPa. The pH setpoint value was 7.5, and pH ranged from 7.0 to 7.8 (note that the culture reached values higher than 7.8 until CO₂ had a homogenous contact with the medium culture). This pH set point of 7.5 was selected based on previous literature that reported a pH preference of *Nostoc* sp. ranging from 7 to 8 [33].

After the stock photobioreactor reached a minimum biomass concentration of 0.15 g VSS L⁻¹, the biomass was considered ready for inoculation.

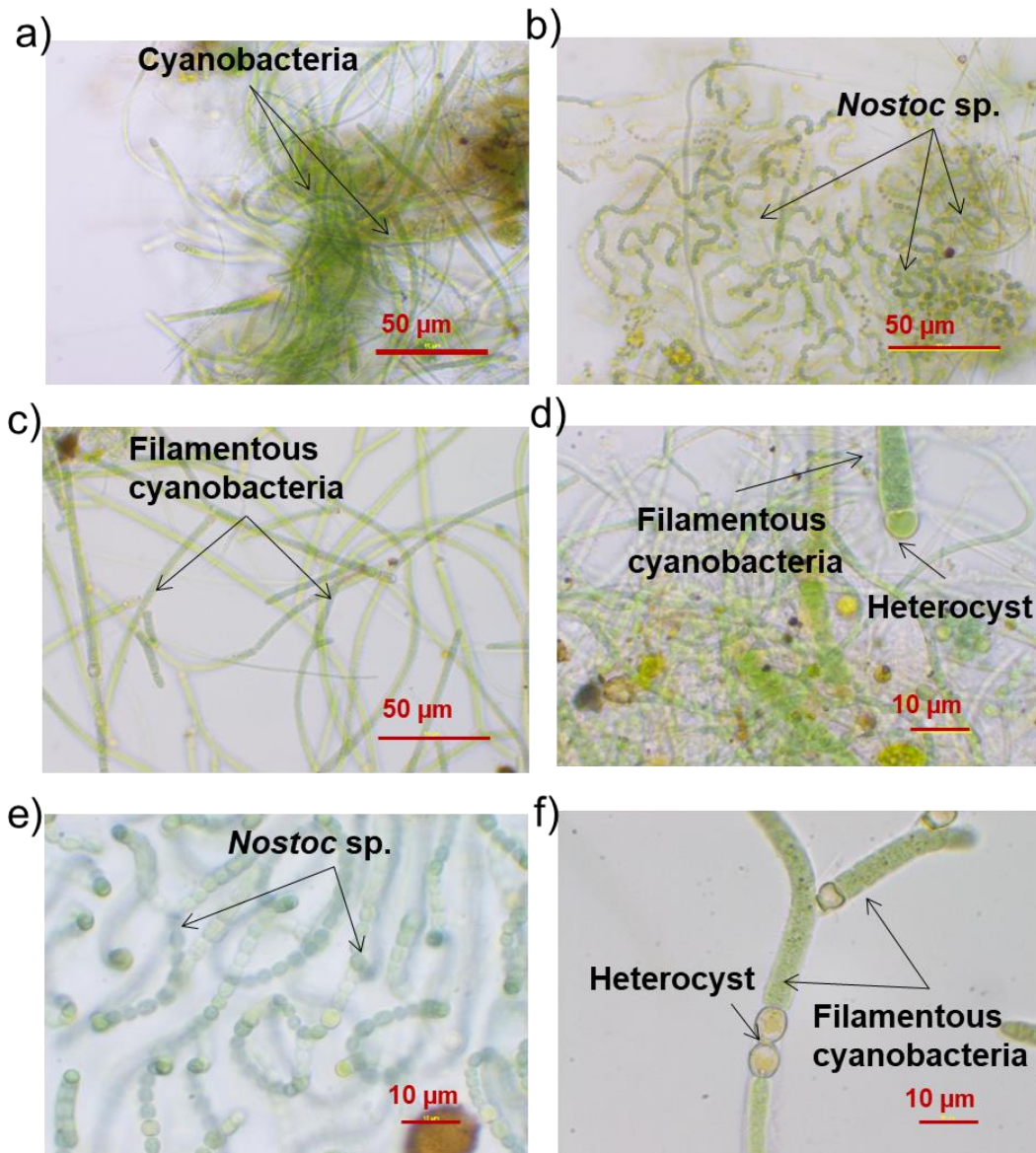


Fig. 1 Microscopic images illustrating the microbial diversity of biomass maintained in the stock photobioreactor; a), b), c) general view of the culture observed at 500x; d) detail of the lateral side of a filamentous cyanobacterial floc; e) detail of a floc of species resembling *Nostoc* sp.; f) detail of filamentous cyanobacteria with heterocyst. All the observations were performed in bright field microscopy.

2.2 Experimental set-up

Lab-scale experiments were performed in two periods carried out in two consecutive months, namely Period 1, carried out during October for 30 days and period 2, carried out

during November for 25 days. Experiments were carried out in different lab-scale closed polymethacrylate photobioreactors with a total volume of 3 L and a working volume of 2.5 L. The photobioreactors were operated in semi-continuous mode using municipal wastewater as feeding. The municipal wastewater used in this experiment was the effluent from a primary settler with a useful volume of 7 L, a surface area of 0.03 m², hydraulic retention time (HRT) between 0.7–1.4 h, and a hydraulic surface loading rate of 11.3 m³/m²·d⁻¹ treating raw urban wastewater from a nearby municipal sewer [37]. Wastewater influent characteristics of the two periods were measured according to the methods described in section 2.3 and are summarized in Table 1.

Four photobioreactors, namely A_F, A_D, B_F, and C_F, were operated in a semi-continuous mode according to the conditions shown in Table 2. During period 1, two photobioreactors, namely A_F and A_D, operated simultaneously at a HRT of 10 days. This operation means that 0.25 L of mixed liquor were removed once a day and 0.25 L of wastewater were subsequently added. The A_F photobioreactor was operated with undiluted wastewater, whereas the A_D was operated with the wastewater diluted with deionized water in a ratio of 2:1. After finishing period 1, B_F and C_F photobioreactors started operation simultaneously and were fed with undiluted wastewater. B_F operated at 8 days of HRT by daily withdrawing approximately 0.313 L of the mixed liquor, and subsequently, replacing this volume with wastewater. Whereas photobioreactor C_F, operated at 6 days of HRT by removing approximately 0.416 L of the mixed liquor, and subsequently, replacing this volume with undiluted wastewater.

All the photobioreactors started with the inoculation of 0.2 L of settled and thickened biomass obtained from the stock photobioreactor described in Section 2.1. To obtain this biomass, 3 L of mixed liquor were taken and let rest in Imhoff cones for 30 min. Biomass

settled by gravity was then collected. Once inoculated, biomass was re-suspended with approximately 2.3 L nitrogen-free BG-11 growth medium. The growth medium was composed of: 0.04 g L⁻¹ K₂HPO₄, 0.072 g L⁻¹ CaCl₂·2H₂O, 0.001 g L⁻¹ Na₂EDTA, 0.075 g L⁻¹ MgSO₄·7H₂O, 0.006 g L⁻¹ C₆H₈FeNO₇ (ammonium ferric citrate), 0.006 g L⁻¹ C₆H₈O₇ (citric acid), and trace elements: 0.00286 g L⁻¹ H₃BO₃, 0.00039 g L⁻¹ Na₂MoO₄·2H₂O, 0.0018 g L⁻¹ MnCl₂·4H₂O, 0.00008 g L⁻¹ CuSO₄·5H₂O, 0.00022 g L⁻¹ ZnSO₄·7H₂O and 0.00005 g L⁻¹ Co(NO₃)₂·6H₂O [38]. As explained before, all the photobioreactors operated in semi-continuous feeding, which means that the system works as the batch feeding, but with daily dilutions (renewed fraction). The proportions of growth medium replaced by wastewater in each photobioreactor is explained in Appendix A2-A and Excel file A2-B. It should be noticed that all the experiments lasted during at least three hydraulic retention time after the adaptation phase.

Photobioreactors A_F, A_D, B_F, and C_F set-up was performed using the configuration described in [12]. Briefly, all the photobioreactors maintained in alternate light: dark phases of 12 h. Illumination during the light phase consisted of two external halogen lamps (60 W) placed at opposite sides of each photobioreactor and providing 220 μmol m⁻² s⁻¹ of light. Due to the lamp type (warm light), culture temperature was highly influenced during the light phase, reaching up to 35 °C, for this reason the temperature was continuously measured by a probe inserted in the photobioreactors (ABRA, Canada) and kept constant at 27 (±2) °C through a water jacket around the photobioreactors. This temperature decreased 1 to 2 °C during the dark cycle. It should be highlighted that soil *Nostoc* sp. species can tolerate high variations of temperatures (from 7 to 45°C) [30,39]. Photobioreactors were continuously agitated with a magnetic stirrer set at 250 rpm. The pH was continuously monitored with a pH sensor

(HI1001, HANNA, USA) and kept at 7.5 with a pH controller (HI 8711, HANNA, USA) by the automated addition of HCl 0.1 N or NaOH 0.1 N (depending on pH). Mixed liquor discharging and feeding was performed each day at the end of the dark phase by the automatic addition/withdrawal accomplished by peristaltic pumps (Appendix A3).

Table 1. Average (standard deviation) of the main quality parameters of the municipal wastewater used as influent and composition of BG-11 growth medium used during the inoculum.

Parameter	Period 1 ^{a c}	Period 2 ^b	BG-11
pH (n=30/24)	7.2 (0.5)	7.4 (0.6)	7
TSS [g·L ⁻¹] (n=6/8)	0.31 (0.17)	0.17 (0.07)	-
VSS [g·L ⁻¹] (n=6/8)	0.29 (0.14)	0.15 (0.08)	-
TOC [mg·L ⁻¹] (n=8/9)	179.28 (61.54)	121.54 (24.04)	-
SOC [mg·L ⁻¹] (n=8/9)	31.49 (18.11)	51.17 (16.99)	-
TIC [mg·L ⁻¹] (n=8/9)	71.46 (12.25)	85.09 (5.82)	4.25
TAN [mg·L ⁻¹] (n=10/11)	35.44 (11.02)	46.26 (6.97)	0.32
N-NO ₃ ⁻ [mg·L ⁻¹] (n=10/12)	<LOD	<LOD	0.006
N-NO ₂ ⁻ [mg·L ⁻¹] (n=10/12)	<LOD	<LOD	-
TIN [mg·L ⁻¹] (n=10/12)	35.44 (11.02)	46.26 (6.97)	4.55
TON [mg·L ⁻¹] (n=10/12)	35.27 (17.33)	20.48 (9.19)	-
TN [mg·L ⁻¹] (n=10/12)	70.71 (14.47)	66.74 (6.70)	4.55
IP [mg·L ⁻¹] (n=10/12)	2.91 (0.39)	2.97 (0.33)	7.11
TOP [mg·L ⁻¹] (n=10/12)	5.69 (3.09)	2.25 (1.31)	-
TP [mg·L ⁻¹] (n=10/12)	8.59 (3.07)	5.22 (1.41)	7.11

LOD: Limit of detection (0.05 mg·L⁻¹).

^a Samples were taken during 30 days.

^b Samples were taken during 25 days.

^c Municipal wastewater used in one of the photobioreactors of this period (A_D) was diluted in a ratio 2:1 with deionized water.

TSS, total suspended solids; VSS, volatile suspended solids; TOC, total organic carbon; SOC, soluble organic carbon; TIC, total inorganic carbon; TAN, total ammoniacal nitrogen; N-NO₃⁻, nitrate; N-NO₂⁻, nitrite; TIN, total inorganic nitrogen; TON, total organic nitrogen; TN, total nitrogen; IP, inorganic phosphorus; TOP, total organic phosphorus; TP, total phosphorus.

Table 2. Type of operation and average (standard deviation) of volumetric loads (VL) applied to the photobioreactors used in this study (n=8-12).

Period	Photobioreactor	HRT [d ⁻¹]	VL-TOC [mg·L ⁻¹ ·d ⁻¹]	VL-TIC [mg·L ⁻¹ ·d ⁻¹]	VL-TN [mg·L ⁻¹ ·d ⁻¹]	VL-TP [mg·L ⁻¹ ·d ⁻¹]
1	A _F	10	17.93±6.16	7.15±1.25	7.07±1.45	0.86±0.31
	A _D	10	8.96±3.08	3.57 ±0.61	3.54 ±0.72	0.43 ±0.15
2	B _F	8	15.17±3.00	10.62±0.73	8.33±0.84	0.65±0.18
	C _F	6	20.22±4.00	14.16±0.97	11.11±1.11	0.87±0.23

HRT, hydraulic retention time; TOC, total organic carbon; TIC, total inorganic carbon; TN, total nitrogen; TP, total organic phosphorus; TP, total phosphorus. A_F, photobioreactor operated at 10 d of HRT with undiluted wastewater; A_D, photobioreactor operated at 10 d of HRT with diluted wastewater in a ratio 2:1; B_F, photobioreactor operated at 8 d of HRT with undiluted wastewater; C_F, photobioreactor operated at 6 d of HRT with undiluted wastewater.

2.3 Wastewater analysis procedures

Parameters concerning water quality and biomass production were determined in triplicate and analyzed from the influent wastewater, mixed liquor, and effluent (supernatant after settling), equivalent to the mixed liquor of the culture, twice a week at the end of the dark phase. Nutrients analyzed were total ammoniacal nitrogen (TAN), total organic carbon (TOC), total inorganic carbon (TIC) and soluble organic carbon (SOC), inorganic phosphorus (IP), nitrite (N-NO₂⁻), nitrate (N-NO₃⁻), total nitrogen (TN) and total phosphorus (TP). TAN (sum of N-NH₃⁻ and N-NH₄⁺) was determined using the colorimetric method indicated in [40]. Inorganic phosphorus (IP) (measured as orthophosphate P-PO₄³⁻), N-NO₂⁻ and N-NO₃⁻ concentrations were analyzed using an ion chromatograph DIONEX ICS1000 (Thermo-scientific, USA), while TOC, TIC, OC, IC, and TN were analyzed by using a C/N analyzer (21005, Analytikjena, Germany). Total phosphorus (TP) was analyzed following the methodology described in [41]. Total inorganic nitrogen (TIN) was the sum of N-NO₂⁻, N-NO₃⁻, and TAN. Total organic nitrogen (TON) (in dissolved and particulate form) was measured in the effluent supernatant and calculated as the difference between TN and TIN.

Also, total organic phosphorus (TOP) was measured in the effluent supernatant and determined as the difference between TP and IP.

Removal efficiencies for TN, TIN, TAN, IP, TP, TOC and TIC in (%) were calculated according the following formula:

$$X (\% \text{ removal}) = \frac{X \text{ influent} - X \text{ effluent}}{X \text{ influent}} * 100$$

Where X is the corresponding parameter concentration.

Total suspended solids (TSS) and volatile suspended solids (VSS) were measured in the mixed liquor three days per week, while the Chlorophyll *a* was analyzed twice a week. These parameters were analyzed using procedures described in [41].

Dissolved oxygen (DO) was measured directly in each photobioreactor, inserting a sensor with a dissolved oxygen-meter (EcoScan DO 6, Thermo-scientific, USA) in the mixed liquor during the second half of the light phase.

2.4 Microbial characterization

Quantitative analysis of microalgae and cyanobacteria was performed by microscopic area cells counting three times per week [42]. To this aim, 20 µL of mixed liquor was added to a slide with a coverslip, and individual cells were counted per field until reaching ~400 cells to have a total standard error lower than 5% [43], alternating bright field microscopy and fluorescence microscopy. Eukaryotic microalgae were identified and quantified by morphology in bright field microscopy at 40X, while cyanobacteria species were identified and counted using fluorescence microscopy with the operation of filters containing an

excitation filter (510-560 nm), emission filter (590 nm) and a *dichroic* beam splitter (575 nm), using phycobiliproteins fluorescence in cyanobacteria as an identification marker [44]. Both bright field and fluorescence microscopy were performed using a fluorescence microscope (Eclipse E200, Nikon, Japan).

The population changes of microalgae species within each photobioreactor were monitored by microscopy once a week. Microbial visualization was performed in an optic microscope (Motic, China) equipped with a camera (Fi2, Nikon, Japan) connected to a computer (software NIS-Element viewer®). Cyanobacteria and microalgae species were identified *in vivo* using identification keys from conventional taxonomic books [45,46], as well as a database of cyanobacteria genus [31,47].

2.5 Carbohydrate quantification

Carbohydrate content was measured twice per week in all the photobioreactors at the end of the dark phase. For this purpose, 50 mL of mixed liquor were collected and centrifuged (4200 rpm, 10 min), frozen at -80°C overnight in an ultra-freezer (Arctiko, Denmark) and finally freeze-dried under 24 h in a lyophilizer (-110°C , 0.049 hPa) (Scanvac, Denmark). Carbohydrates extraction and quantification were conducted using the methodology described in [48], measuring therefore total sugars and related substances. Carbohydrate content percentage of VSS was calculated according to [49], following the formula:

$$\% \text{Carbohydrates} = \frac{\text{gCarbohydrates}}{\text{gVSS}} * 100$$

2.6 Statistical analysis

Data of the main nutrients (TN, TP, TOC, TIC), biomass concentration (VSS), chlorophyll *a*, and carbohydrate production were examined to determine whether they met normality assumptions using a Shapiro-Wilks test and were transformed (log10 or square root) when necessary. All statistical analyses were performed using Minitab 18 (Minitab Inc., PA, USA). Normal data were then submitted to a one-way analysis of variance (ANOVA) with repeated measured. When the experimental data presented a significant difference, then F-test was conducted at a 95% confidence interval level.

3. Results and discussion

3.1 Wastewater treatment

Throughout the experimental time, the municipal wastewater used as influent in the two periods showed similar characteristics in pH, TN, and IP (Table 1). Wastewater used in period 1 showed higher TSS, TOC and TON, with higher variability than in period 2. Otherwise, period 2 showed higher TIN and doubled the TIN: TON ratio (2:1) in comparison with period 1 (1:1). The variability of the wastewater caused such differences between influents. Wastewater treatment performance of the photobioreactors was evaluated in terms of organic carbon, nitrogen, and phosphorus removals. Photobioreactor's liquor can be considered as the effluent of the wastewater treatment and its characteristics are summarized in Table 3. Average removals are presented in Appendix A4.

In general, A_F and A_D averaged 86 and 82% in the removal of TOC, respectively. Both photobioreactors showed better TOC removal in the first fifteen days of operation. Subsequently, TOC concentration in the photobioreactors increased following the increment in influent TOC (Fig. 2). Besides, the TIC content was almost completely removed (>93%) during all the experimental time. Regarding B_F and C_F , both systems showed higher removal

efficiencies than the photobioreactors operated at 10 d HRT, removing TOC by 89% (Fig. 2). While TIC was almost completely consumed (>96%) in both photobioreactors, these high removals in TIC suggest a carbon limitation in the cultures. ANOVA test of TOC results in all the experiments did not show significant differences ($P>0.05$). However, F-test comparison of TIC data showed statistical differences between the periods but without significant differences of A_F against A_D and B_F against C_F .

Concerning nitrogen species, A_F and A_D removed over 98% of TAN and TON, while the total removal of TN in these photobioreactors were 95 and 98%, respectively, without significant differences ($P>0.05$) (Fig. 3a and 3b). As most of the time low values of $N-NO_2^-$ and $N-NO_3^-$ were obtained (Table 3), it is assumed that eukaryotic microalgae/cyanobacteria consumed all the TAN and TON (mineralized to TAN). Otherwise, B_F and C_F showed very low concentrations ($<2 \text{ mg L}^{-1}$) of $N-NO_2^-$, $N-NO_3^-$ (thus, low nitrification) and TON, while presenting TAN accumulation until achieving over 20 mg L^{-1} in the last two weeks of operation (Fig. 3c and 3d). This increase in TAN concentration likely occurred due to the carbon limitation occurring during period 2. Despite those increments in TAN concentrations, the general TN removal in B_F and C_F was 75% and 66%, respectively. It should be noticed that TN values in these last photobioreactors did not show significant differences, but it was significantly lower than A_F and A_D ($P=0.03$).

As already mentioned, the photobioreactors were firstly inoculated with nitrogen-free BG-11 growth medium containing phosphorus. Hence, the prolonged nitrogen limitation caused an initial concentration of approximately 5 mg L^{-1} of IP in the cultures. In A_F , the general TP removal averaged 53% (Fig. 3), whereas high values of IP in the effluent during the experiment resulted in removal efficiency of only 8%, while TOP was removed by 82%.

Similarly, A_D maintained TP concentration mostly in the form of IP around 2.5 mg L⁻¹ (Fig. 3). This photobioreactor achieved 55% of IP and 91% of TOP removal efficiency, while the general TP removal was 78%. On the other hand, B_F presented high IP concentrations despite that the influent contained a lower TP load than in A_F (Table 2). Until day 20, TP (almost composed by IP) maintained concentrations closed to the influent (5.22±1.41 mg L⁻¹) (Fig.3). Notwithstanding, the average removal percentages were TP 62%, IP 57% and, TOP 65%. The photobioreactor C_F presented a lower removal than the photobioreactor aforementioned due to the higher load applied (Fig. 3). IP was removed by 7%, TOP by 56% and TP by 59%. ANOVA test of TP did not show significant differences among all the photobioreactors.

Overall, the general performance of all the systems presented in this study showed high organic matter removal efficiencies even under different operational conditions (i.e., HRT, loads), especially in terms of TOC (Table 3). This is indicative of heterotrophic bacteria present in the cultures. Considering the photobioreactors used as wastewater treatment systems, results obtained agreed with other lab and pilot studies that focused on microalgae biomass production in photobioreactors using municipal wastewater effluents as nutrient sources [37,50–53]. Comparably, nutrients removal is similar to previously studies conducted in open ponds using mixed cultures mainly dominated by *Chlorella* sp. under relative HRT in continuous operation (>63% N and >11% P removal) [52,53].

Table 3. Average (standard deviation) of the main quality parameters of the effluent (supernatant after biomass settling) and mixed culture characteristics of the cultures during the experiment.

Parameter	A _F	A _D	B _F	C _F
pH ^a (n=30/30/24/21)	7.56 (0.11)	7.82 (0.14)	7.32 (0.05)	7.36 (0.16)
OD ^a (n=30/30/24/21)	6.03 (0.28)	7.01 (0.51)	7.59 (1.81)	10.34 (10.35)

TSS [g·L ⁻¹] ^a (n=30/30/24/21)	0.51 (0.21)	0.40 (0.14)	0.54 (0.14)	0.32 (0.05)
VSS [g·L ⁻¹] ^a	0.44 (0.20)	0.38 (0.19)	0.46 (0.06)	0.31 (0.05)
Biomass production [g·L ⁻¹ d ⁻¹] ^a	0.04 (0.02)	0.04 (0.01)	0.06 (0.01)	0.05 (0.02)
TOC [mg·L ⁻¹] ^a (n=6/6/14/14)	256 (140)	258 (112)	244 (40)	187 (56)
SOC [mg·L ⁻¹] (n=6/6/14/14)	25.5 (11.5)	31.7 (14.5)	13.5 (3.3)	13.5 (2.8)
TIC [mg·L ⁻¹] (n=6/6/14/14)	5.17 (1.88)	3.81 (0.94)	3.15 (2.22)	3.96 (4.03)
Chlorophyll <i>a</i> [mg·L ⁻¹] ^a	1.13 (0.58)	1.10 (0.66)	0.69 (0.32)	0.53 (0.19)
TAN [mg·L ⁻¹] (n=9/9/11/11)	0.68 (1.01)	0.71 (1.07)	16.09 (10.32)	17.92 (9.55)
N-NO ₃ ⁻ [mg·L ⁻¹] (n=9/9/11/11)	0.12 (0.31)	0.08 (0.25)	0.15 (0.35)	0.38 (0.54)
N-NO ₂ ⁻ [mg·L ⁻¹] (n=9/9/11/11)	2.25 (3.17)	0.42 (1.18)	<LOD	1.32 (1.85)
TIN [mg·L ⁻¹] (n=9/9/11/11)	3.05 (4.49)	1.21 (2.50)	16.24 (10.65)	19.62 (12.94)
TON [mg·L ⁻¹] (n=9/9/11/11)	<LOD	<LOD	0.05 (0.16)	2.57 (3.70)
TN [mg·L ⁻¹] (n=9/9/11/11)	3.05 (4.49)	1.21 (2.50)	22.19 (8.76)	16.29 (9.56)
IP [mg·L ⁻¹] (n=11/11/12/12)	2.67 (1.38)	1.31 (1.20)	1.29 (0.47)	2.75 (1.47)
TOP [mg·L ⁻¹] (n=11/11/12/12)	0.74 (0.83)	0.37 (0.83)	0.79 (1.23)	0.98 (1.88)
TP [mg·L ⁻¹] (n=11/11/12/12)	3.05 (4.49)	1.68 (2.03)	3.09 (1.70)	3.73 (3.35)

LOD: Limit of detection (0.05 mg L⁻¹).

^a Parameters measured in the mixed liquor.

A_F, photobioreactor operated at 10 d of HRT with undiluted wastewater; A_D, photobioreactor operated at 10 d of HRT with diluted wastewater in a ratio 2:1; B_F, photobioreactor operated at 8 d of HRT with undiluted wastewater; C_F, photobioreactor operated at 6 d of HRT with undiluted wastewater; TSS, total suspended solids; VSS, volatile suspended solids; TOC, total organic carbon; SOC, soluble organic carbon; TIC, total inorganic carbon; TAN, total ammoniacal nitrogen; N-NO₃⁻, nitrate; N-NO₂⁻, nitrite; TIN, total inorganic nitrogen; TON, total organic nitrogen; TN, total nitrogen; IP, inorganic phosphorus; TOP, total organic phosphorus; TP, total phosphorus.

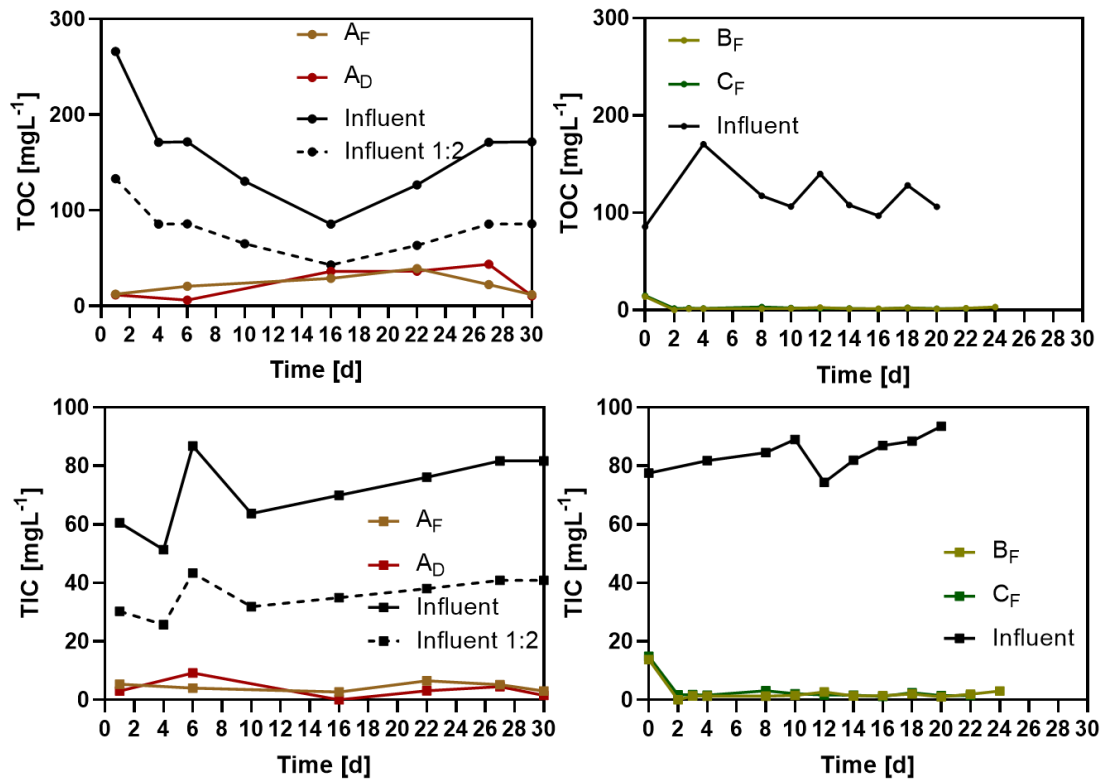


Fig. 2. Time course of influent and photobioreactor effluent (supernatant after biomass settling) during the experiment in period 1 (a) and period 2 (b). Total organic carbon (TOC) (above), total inorganic carbon (TIC) (below) in A_F (photobioreactor operated at 10 d of HRT with undiluted wastewater); A_D (photobioreactor operated at 10 d of HRT with diluted wastewater in a ratio 2:1); B_F (photobioreactor operated at 8 d of HRT with undiluted wastewater); and C_F (photobioreactor operated at 6 d of HRT with undiluted wastewater).

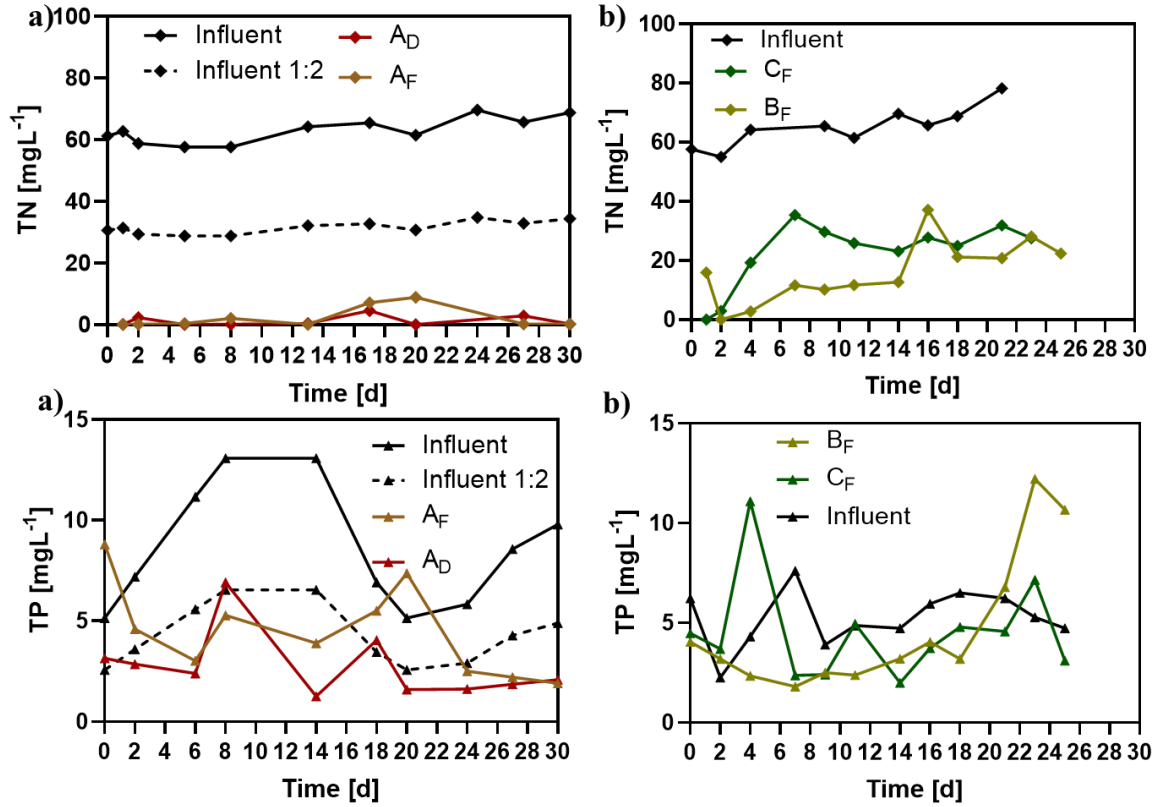


Fig. 3. Time course of influent a) period 1 and b) period 2, and photobioreactor effluent (supernatant after biomass settling) total nitrogen (TN) (up) and total phosphorus (TP) (down) in A_F, photobioreactor operated at 10 d of HRT with undiluted wastewater; A_D, photobioreactor operated at 10 d of HRT with diluted wastewater in a ratio 2:1; B_F, photobioreactor operated at 8 d of HRT with undiluted wastewater; C_F, photobioreactor operated at 6 d of HRT with undiluted wastewater.

3.2 Biomass production and composition

Notwithstanding the relatively high nitrogen removal observed in A_F and A_D, the biomass concentration steadily increased throughout the experiment. The A_F and A_D photobioreactors reached a stable biomass concentration with a variation from 0.55 g·L⁻¹ to 0.75 g·L⁻¹ between days 20 and 30. Consequently, the biomass production followed the same trend, with an average value of 0.07 g·L⁻¹·d⁻¹ in A_F and 0.05 g·L⁻¹·d⁻¹ in A_D from day 20 to day 30. A similar trend was observed in chlorophyll *a* content, thus, after day 15, chlorophyll *a* reached and maintained value of 1.55 mg·L⁻¹ (A_F) and 1.58 mg·L⁻¹ (A_D) (Fig. 4). Both

photoreactors did not present any significant difference between them ($P>0.05$) in biomass production and chlorophyll *a*.

Biomass production in C_F and B_F showed different trends in comparison to A_F and A_D . In the case of C_F , it showed a decreasing pattern over time, achieving a constant biomass production of $0.06\pm0.01 \text{ g}\cdot\text{L}^{-1}\text{d}^{-1}$ between day 8 and the end of the experiments. In contrast, biomass production in B_F decreased in the first days of operation until day 8, and subsequently, the culture showed values of $0.04\pm0.02 \text{ g}\cdot\text{L}^{-1}\cdot\text{d}^{-1}$ (Fig, 4). Despite the different tendencies between them, F-test did not find significant differences ($P>0.05$). Conversely, they showed significant differences against A_F and A_D , which can be attributed to the fact that photobioreactors of B_F and C_F did not present nitrogen limitation but carbon limitation. In the case of chlorophyll *a* concentration in B_F , despite showing some pick in different days, it was observed a decreasing trend along with the experiment. Similarly, C_F presents this same pattern more clearly ($P=0.014$), decreasing from 0.6 mg L^{-1} to an average of 0.4 mg L^{-1} from day 15 to the last day of the experiment.

The fact that biomass increased in A_F and A_D despite the nitrogen limitation can be attributed to two reasons: the first one comprises the characteristics of the culture. Both photobioreactors started with a dominant culture of N-fixing cyanobacteria, thus cyanobacteria with heterocyst can grow with reduced nitrogen content because of their ability to uptake atmospheric nitrogen [2,4,54]. The second reason is related to the non-N-fixing cyanobacteria; in this case, the nitrogen introduced to the photobioreactor might be enough to promote microalgae grow in the first half of the experiment. Afterward, the nitrogen in the culture might not be enough, and cells could consume their stored nitrogen forms. This same pattern was observed in a batch-test study [12], where cyanobacteria were submitted to a

nitrogen-depleted growth medium, biomass maintained an increasing trending during 8 days, subsequently, biomass remained stable until day 15. In that study, the complete unavailability of nitrogen led to pigment damage (chlorosis) [55]. It should be notice that in this work, A_F and A_D did not present chlorosis and chlorophyll *a* increased, which implies that nitrogen load could have been enough to maintain the growth without using intracellular nitrogen.

Nonetheless, B_F and C_F were affected by carbon limitation, resulting in limited biomass growth and deficient nutrients removal. Likewise, in [56], a culture dominated by cyanobacteria fed with growth medium, showed better responses in nutrients removal and biomass growth with nitrogen and phosphorus limitation than with carbon limitation, once the culture lacked carbon, biomass maintained the same concentration along the time.

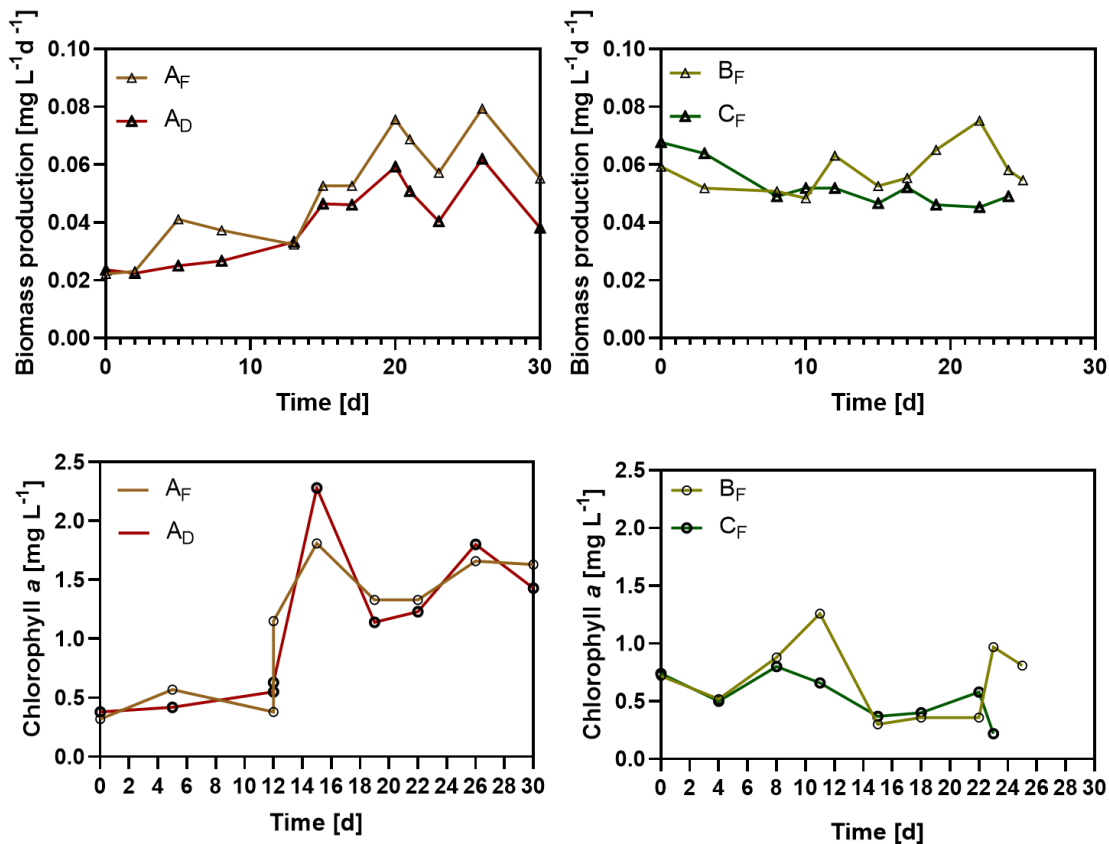


Fig. 4. Time course of biomass production (above) and chlorophyll *a* (below) content in A_F, photobioreactor operated at 10 d of HRT with undiluted wastewater; A_D, photobioreactor operated at 10 d of HRT with diluted wastewater in a ratio 2:1; B_F, photobioreactor operated at 8 d of HRT with undiluted wastewater; C_F, photobioreactor operated at 6 d of HRT with undiluted wastewater.

3.3 Biomass composition

Since all the cultures were non-controlled and operated under non-sterile conditions, contamination by other microorganisms was expected during the experiment. All the cultures had cyanobacteria dominance by 99.8% at the beginning of the experiments, but a certain increase of other microorganisms was detected, depending on the HRT and nutrients availability in the cultures. However, cyanobacteria dominance was maintained during the experimental time (>98%), and only the photobioreactors with lower HRT (B_F and C_F) showed a slightly higher incidence of other microorganisms such as green algae, diatoms, and protozoa (Fig. 5). Statistical treatment of all the data showed that cyanobacteria content was not significantly different between experiments ($p>0.05$), but it presents significant differences against other microorganisms and treatments. Otherwise, F-tests performed to green algae, diatoms and protozoa content were not significantly different among each other, except for protozoa content in C_F ($P=0.021$).

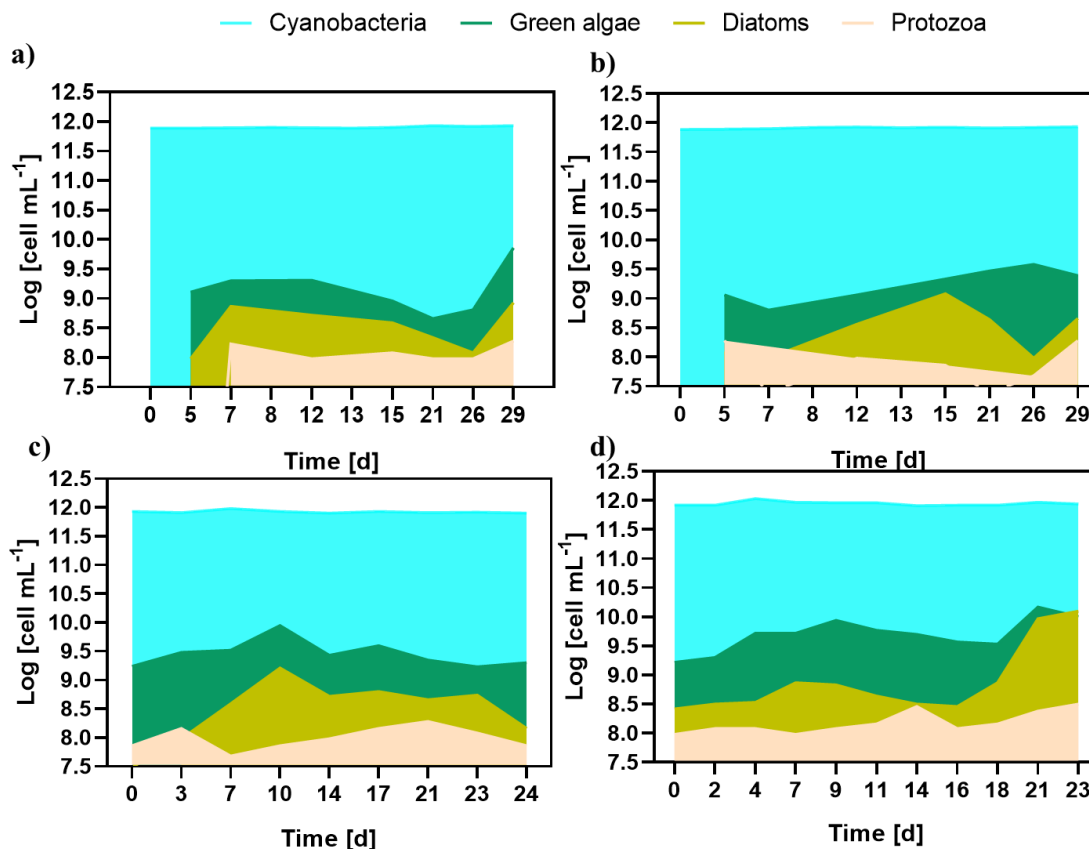


Fig. 5 Biomass composition in a) A_F, photobioreactor operated at 10 d of HRT with undiluted wastewater; b) A_D, photobioreactor operated at 10 d of HRT with diluted wastewater in a ratio 2:1; c) B_F, photobioreactor operated at 8 d of HRT with undiluted wastewater; d) C_F, photobioreactor operated at 6 d of HRT with undiluted wastewater.

Microbial observations during the experiments showed that filamentous cyanobacterial species were dominating the culture over time, a table showing species groups abundance is stated in Appendix A5-A8. A_F revealed intense domination by those species grouped in large flocs and presented the appearance of colonial green algae species and diatoms in the first days of operation. After day 6, species inoculated in this photobioreactor, such as *Nostoc*-like species and branched filamentous cyanobacteria, were barely observed. Interestingly, the culture was increasingly dominated by filamentous cyanobacteria species (Appendix A9). Conversely, A_D showed a higher presence of *Nostoc*-like species, in elongate spherical mucilage aggregations during the experimental time. In Appendix A10, filamentous

species with heterocyst were fragmented and usually observed in elongate spherical mucilage aggregations. Along with this type of species, filamentous cyanobacteria were also observed together with some colonies of green algae and diatoms.

With B_F, a strong presence of heteropolar cyanobacteria resembling *Calothrix* sp. along with fragmented *Nostoc*-like species was observed since the day three of operation (Appendix A11). At the end of the experiment, the culture was dominated by both non-N-fixing filamentous cyanobacteria (without heterocytes) and N-fixing filamentous cyanobacteria (with heterocytes). Similarly, C_F also presented a dominant presence of heteropolar cyanobacteria resembling *Calothrix* sp., with colonial coccoid cyanobacteria in elongate spherical mucilage aggregations. Colonial green algae along with the diatom resembling *Nitzschia* sp. were also observed along the experimental time (Appendix A12). B_F and C_F showed increasing concentrations of in green algae and other microorganisms than cyanobacteria, probably because of the high nutrients and carbon loads in the culture combined with low HRT, providing conditions for other microorganisms to compete and grow. This trend was observed also in previous studies [57,58] growing mixed cultures in wastewater. While cyanobacteria dominance was observed in cultures submitted to high HRT [59].

The fact that non-heterocyst filamentous cyanobacteria rapidly invaded the initial population dominated by cyanobacteria with heterocyst can be attributed to the availability of ammoniacal nitrogen in the culture. Furthermore, the *Nostoc*-like species population decreases and morphological changes can be explained by the fragmentation of trichomes and destruction of heterocyst occurring when cells can reduce nitrogen-fixing activity, in this case, as consequence of ammoniacal availability [60]. In the case of spherical mucilage

aggregations by *Nostoc* sp., some studies have shown that this occurs as a mechanism to prevent grazing and increases longevity and nutrient recirculation [39].

It is worth noting that all the cultures presented good natural sedimentation since filamentous cyanobacteria species are recognized as easy-settling species [59,61]. Besides, an efficient separation requires the presence of biomass filamentous species forming big colonies, while coccal green algae species were observed inside big flocs with cyanobacteria. Only C_F showed low turbidity due to the presence of *Chlorella* sp. that dispersed out the flocs. The efficient sedimentation of this system is an important issue since it can contribute to the economic feasibility of the process. Indeed, harvesting is an important issue, that account for 20–30% of the cost of the process [56,62].

3.4 Carbohydrate production

Two clear patterns in total carbohydrate accumulation were observed during the experiments (Fig. 6). Photobioreactors presented different initial carbohydrate content according to the period in which they were inoculated. As explained in section 2.2, the biomass was maintained during approximately one month in a stock photobioreactor under N-limited conditions before photobioreactor's inoculation. The A_F and A_D photobioreactors were firstly inoculated (period 1) while B_F and C_F started their operation 30 days after (period 2). Thus, the cultures operated in period 1 showed initial carbohydrate content of 15% in terms of dry cell weight (dcw), whereas B_F and C_F contained an intracellular content of 30% on the day of inoculation.

In the case of A_F and A_D, both cultures showed first a slight concentration decrease from 15% to 5% and 10%, respectively, in the first days of operation. Subsequently, A_F and A_D carbohydrate intracellular contents gradually increased reaching 34% and 48%, on the

last day of operation, respectively. Statistical analysis showed that carbohydrate concentration in both photobioreactors was not significantly different $P>0.05$ (Fig. 6). These results suggest that the best condition to achieve the highest carbohydrate concentration was the HRT of 10 days and nutrients and carbon loads below the values tested in A_F. This high content of carbohydrates at the end of the experiment can be attributed to the exopolysaccharides mass around the flocs observed in microscopic images of day 30, shown in Appendix A9 and A10. This was a consequence of nutrients availability previously described in section 3.1. Increasing carbohydrates in A_F and A_D can be associated with nitrogen limitation occurring during most of the experimental time (Fig. 2).

According to [63], exopolysaccharides production in soil cyanobacteria occurs only when carbon exceeds the amount of nitrogen available; in this case, exopolysaccharides are a sink for excess fixed carbon. Likewise, accumulation percentages reached in both cultures during a continuous operation are closed to previous studies performed in a two-stage process, cultivation stage and further accumulation stage in long batch operation testing nitrogen limitation in *Synechocystis* sp. [64], *Spirulina maxima* (70, 23% dcw) [65] and *Arthrospira* sp. (65% dcw) [66,67]. These results suggest that nitrogen limitation led to a speedy carbon transformation to carbohydrates in the species of cyanobacteria used in the studies. However, it should be highlighted that those studies were carried out in synthetic growth medium under a 24 h light cycle, while in the present study, municipal wastewater was the only source of carbon, and nutrients and the experiments were performed under a circadian cycle.

The B_F and C_F halved the initial intracellular content of 30% during the first four days of operation. B_F continued a decreasing pattern reaching 14% on day 25 while C_F maintained

a constant carbohydrate content of approximately 12%. F-test performed with both photobioreactors showed that B_F and C_F were not significantly different between each other in terms of carbohydrates concentration ($P>0.05$). However, they present significant differences against A_F and A_D (Fig. 6). Decreasing trending observed in B_F and C_F might be due to the availability of nutrients along with carbon limitation observed from the first days of operation (Fig. 3). Several studies have concluded that carbohydrates accumulation is possible due to the nutrient's limitation or depletion in the culture [13,64,65]. Thus, once that nutrients are available, stored carbon may be rapidly consumed to perform metabolic pathways in the cell as for biomass growth or maintenance [68]. This behavior has also observed in previous studies in sequencing batch photobioreactors [56]. However, carbohydrate content obtained in these two cultures is comparable to the one found in microalgal systems for wastewater treatment. For example, Arcila and Buitrón, [61], obtained 12 and 16% in high rate algal ponds treating urban wastewater operated at hydraulic and solids retention times of 2 and 6 d, respectively. Other studies [58,69] also reached maximum percentages around 12% in a sequencing batch photobioreactor treating secondary effluents and digestate.

The comparison of carbohydrate content and nitrogen availability in all the cultures led to conclude that soil crusts can be used for carbohydrate-enriched biomass production in a wastewater treatment process. Carbohydrate content was conditioned to nutrients and carbon loads applied during the process, HRT of 10 days and nutrients and carbon loads of $17.93\pm 6.16 \text{ mg}\cdot\text{L}^{-1}\cdot\text{d}^{-1}$ TOC; $7.15\pm 1.25 \text{ mg}\cdot\text{L}^{-1}\cdot\text{d}^{-1}$ TIC; $7.07\pm 1.45 \text{ mg}\cdot\text{L}^{-1}\cdot\text{d}^{-1}$ TN and $0.86\pm 0.31 \text{ mg}\cdot\text{L}^{-1}\cdot\text{d}^{-1}$ TP, led to a maximum carbohydrate production of 48% dcw in 30 days of operation without any external source of carbon as CO_2 or NaHCO_3 . Overall, this study

demonstrates that carbohydrate production in a one-stage process of cultivation and accumulation can be improved through the control of hydraulic retention time and nutrients and carbon loads. Further research aiming to test a long-term study is encouraged to determine the maximum carbohydrate content that can be obtained in a continuous process. Additionally, other factors that may affect the culture such as the effect of outdoor conditions (e.g. direct sunlight and temperature) can also be evaluated to scale-up the system and convert the biomass into renewable biofuels.

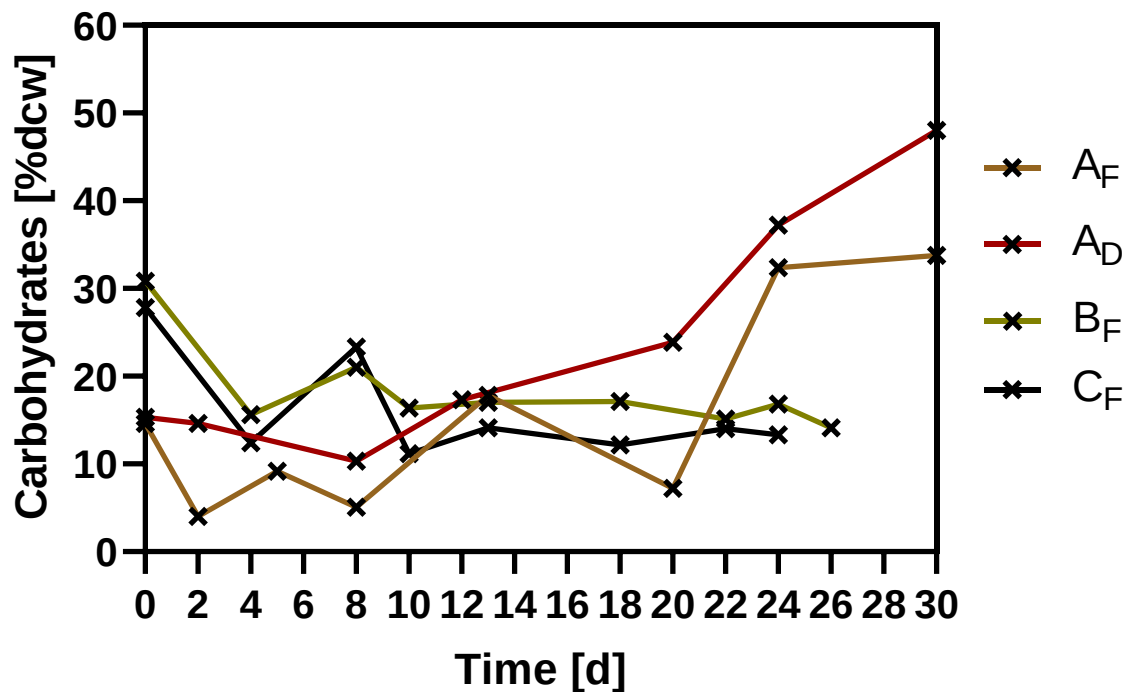


Fig. 6. Accumulation of carbohydrates during the experiment.

4. Conclusions

This work demonstrated that a mixed soil cyanobacterial culture could simultaneously treat municipal wastewater and produce carbohydrates in a one-stage operation. Results

evidenced that high HRT (10 d) and low nutrients load (TN <7.07 and TP<0.86) achieved the highest removals efficiencies in TN >95%, TP 35-78%, TOC >93% and TIC >82%. These high removals led to nitrogen limitation that stimulated a continuous carbohydrates accumulation of up to 48%. Under these conditions, biomass production of 0.05-0.07 mg·L⁻¹·d⁻¹ dominated by easy-settling flocs of N-fixing filamentous cyanobacteria was achieved. Otherwise, lower HRT and thus high nutrients loads promoted carbon limitation, which led to lower nitrogen and phosphorus removals (TN 66-75%, TP 27-58%), poor carbohydrate content (<14%), and low biomass production (0.05-0.07 mg L⁻¹ d⁻¹). Besides the fact that these conditions favored the contamination of green algae species. Overall, this study provides important information about soil cyanobacteria cultivation strategies and further production of valuable by-products from biomass in wastewater treatment systems.

CRediT Author Contributions Statement

D.M. Arias performed the experiments, analysis, interpretation of the data, and drafted the manuscript. E. Uggetti, J. García, A. contributed to the interpretation of the data and critically reviewed the manuscript. All the authors read, edited and approved the manuscript.

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Declaration of conflicting interest

The authors declare that they have no known conflicting financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Statement of Informed Consent, Human/Animal Rights

No conflicts, informed consent, or human or animal rights apply to this study.

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